

## 2.6 Multiplying Binomials With Algebra Tiles

### Lesson Introduction

|                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                        |                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  Benchmark           | MA.912.AR.1.3 Add, subtract, and multiply polynomial expressions with rational number coefficients.                                                                                                                                                                                                                                                                                                                                                   |                                                                        |  Learning Target                                                                                                                                                                                  | <ul style="list-style-type: none"><li>I can multiply binomials using algebra tiles.</li></ul>                                                                                                                                                                                                                                                                                                                    |
|  Benchmark Coherence | Building On<br>MA.8.AR.1.2 Apply properties of operations to multiply two linear expressions with rational coefficients.                                                                                                                                                                                                                                                                                                                              |                                                                        | Working Toward<br>MA.912.AR.1.5 Divide polynomial expressions using long division, synthetic division, or algebraic manipulation.<br>MA.912.AR.1.9 Apply previous understanding of rational number operations to add, subtract, multiply, and divide rational algebraic expressions. |                                                                                                                                                                                                                                                                                                                                                                                                                  |
|  Essential Question  | <ul style="list-style-type: none"><li>How can I represent the multiplication of two binomials using algebra tiles?</li></ul>                                                                                                                                                                                                                                                                                                                          |                                                                        |                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                  |
|  Lesson Narrative    | In this lesson, students will extend their learning of operations with polynomials to multiplying binomials. Students will begin by exploring this concept using the area model. They will then transition into representing this concept using algebra tiles with guided instruction and practice. Students will be given the opportunity to practice and demonstrate understanding of using the algebra tiles to find the product of two binomials. |                                                                        |                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                  |
|  Component Summary | 2.6.1 Warm-Up (~5 min)                                                                                                                                                                                                                                                                                                                                                                                                                                | Complete a math puzzle and describe a pattern used ( <i>SE p. 86</i> ) | 2.6.4 Guided Practice (5-10 min)                                                                                                                                                                                                                                                     | Practice multiplying binomials using algebra tiles ( <i>SE p. 90</i> )                                                                                                                                                                                                                                                                                                                                           |
|                                                                                                       | 2.6.2 Exploration (10 min)                                                                                                                                                                                                                                                                                                                                                                                                                            | Explore multiplication using the area model ( <i>SE p. 86</i> )        | 2.6.5 Your Turn! (5-10 min)                                                                                                                                                                                                                                                          | Use algebra tiles to find the product of two binomials ( <i>SE p. 90</i> )                                                                                                                                                                                                                                                                                                                                       |
|                                                                                                       | 2.6.3 Guided Instruction (15 min)                                                                                                                                                                                                                                                                                                                                                                                                                     | Multiply binomials using algebra tiles ( <i>SE p. 87</i> )             | 2.6.6 Wrap-Up (~5 min)                                                                                                                                                                                                                                                               | Ticket Out the Door on multiplying binomials using algebra tiles ( <i>SE p. 91</i> )                                                                                                                                                                                                                                                                                                                             |
|  Resources         | Glossary                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                        | Materials                                                                                                                                                                                                                                                                            | Additional Preparation                                                                                                                                                                                                                                                                                                                                                                                           |
|                                                                                                       | <u>Key Terms:</u> N/A<br><br><u>Additional Terms:</u> <ul style="list-style-type: none"><li>Binomial</li></ul>                                                                                                                                                                                                                                                                                                                                        |                                                                        | <ul style="list-style-type: none"><li>(Optional) Algebra tiles</li></ul>                                                                                                                                                                                                             | <ul style="list-style-type: none"><li>This lesson is focused on building conceptual understanding of multiplying polynomials through the use of algebra tiles. Consider having physical or virtual algebra tiles prepared for students to use. Also consider how you, as the teacher, can represent the algebra tiles when modeling (using technology, magnetic tiles, paper tiles with magnets, etc.)</li></ul> |

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|----------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <br>Teacher Tips              | Common Misconceptions                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Things to Consider                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|                                                                                                                | <ul style="list-style-type: none"><li>Students may use the incorrect algebra tile with the negative and positive side.</li><li>Student may not connect the area model to the algebra tiles part of the lesson without explicit instruction.</li></ul>                                                                                                                                                                                                                                    | <ul style="list-style-type: none"><li>This lesson is centered around algebra tiles. Students may need additional instruction or support around how to read and use algebra tiles.</li><li>If time is an issue, consider assigning 2.6.5 Your Turn! as homework.</li></ul>                                                                                                                                                                                                                                                               |
|                                                                                                                | Literacy and Language Routines                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Mathematical Thinking and Reasoning (MTR) Standard                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|                                                                                                                | <b>Think-Pair-Share</b><br>2.6.2 Exploration provides students with an opportunity to work with multiplying using the area model. This would be a great opportunity to allow students a few minutes to try it on their own and then pair up with a partner to review their answers and come to a consensus.                                                                                                                                                                              | <b>MA.K12.MTR.2.1: Multiple Representations</b><br>Component 2.6.3 Guided Instruction teaches students how to multiply binomials using algebra tiles. They will connect this learning to the use of the area model that they explored in the 2.6.2 Exploration and extend this representation when learning how to factor in future units. This allows students to express connections between concepts and representations and serves to guide students from concrete to pictorial to abstract representations as learning progresses. |
| <br>Differentiate Instruction | This lesson focuses on the use of algebra tiles as a way to conceptualize multiplying binomials. Consider the use of physical or virtual algebra tiles for this lesson. This can help students get a hands-on experience with the concept. To add physical movement and increase engagement, during 2.6.3 Guided Instruction, consider having students represent algebra tiles and come to the front of the room to represent the algebra tile as you are modeling one of the questions. |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Lesson Notes:                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

## 2.6.1 Warm-Up: Math Puzzle

**Component Overview:** Students will complete a math puzzle to determine the missing value and describe the pattern they used to solve.



### 2.6.1 Warm-Up: Math Puzzle

A math puzzle is shown with one value missing.

|   |   |   |   |
|---|---|---|---|
| 3 | 4 | 6 | 2 |
| 5 | 2 | 3 | 5 |
| 2 | 6 | ? | 3 |
| 1 | 7 | 5 | 7 |



Consider leveraging a Think-Pair-Share routine for this activity. Allow students a minute or two to try and figure out the missing value on their own. Then have them pair up with their partner to share strategies and approaches to come to a consensus on the correct answer. When sharing with the class, have students discuss the methods that they used to look for different approaches to solving the puzzle.

- Find the missing number in the puzzle.

9

- Describe the pattern used to find the missing number.

*For each square, the top two numbers added together equal the bottom two numbers added together.*

## 2.6.2 Exploration: Multiplying With the Area Model

**Component Overview:** Students will explore multiplication through the use of an area model.



### 2.6.2 Exploration: Multiplying with the Area Model

Work with your partner to complete the following.

1. Three multiplication problems are set up using an area model.
  - a. Complete the area models to find each product.

**Model A**

|    |     |     |
|----|-----|-----|
|    | 20  | + 5 |
| 10 | 200 | 50  |
| +5 | 100 | 25  |

**Model A's Product**

375

**Model B**

|    |      |     |
|----|------|-----|
|    | 20   | + 7 |
| 20 | 400  | 140 |
| -5 | -100 | -35 |

**Model B's Product**

405

**Model C**

|    |     |     |
|----|-----|-----|
|    | 20  | -2  |
| 12 | 240 | -24 |
| +3 | 60  | -6  |

**Model C's Product**

270



Students have used the area model before in prior courses but may not recall how to use it. Consider providing a brief review (1 minute) on how an area model works, using different numbers, to jump start the students for this activity, if needed.

This component is intended to be a quicker refresher of the model with known numbers prior to moving into using it with unknown quantities.



Consider asking the following:

- What do the numbers outside the rectangle represent in the area model?
- What do the numbers inside the model represent?
- Do you think you would have still gotten the same answer if the models had been written with different numbers? Why or why not?

## 2.6.2 Exploration: Multiplying With the Area Model (continued)

- b. Match each area model to the multiplication expression it represents.

| Multiplication Expression | Area Model                                                                                      |
|---------------------------|-------------------------------------------------------------------------------------------------|
| $(15)(27)$                | <input type="radio"/> Model A<br><input type="radio"/> Model B<br><input type="radio"/> Model C |
| $(15)(25)$                | <input type="radio"/> Model A<br><input type="radio"/> Model B<br><input type="radio"/> Model C |
| $(15)(18)$                | <input type="radio"/> Model A<br><input type="radio"/> Model B<br><input type="radio"/> Model C |

- c. Complete the statements.

In the area models, each rectangle represents part of

the

- ☐ factor.  
☐ product.

Parts of each

- ☐ factor  
☐ product

are

multiplied to find four partial products. The product of

the two factors is found by

- ☐ adding  
☐ multiplying

the partial

products.



*When reviewing this activity, spend an extra minute or two on explicit use of the language in question 1c. These terms will be important as students transition to the upcoming components in this lesson.*

## 2.6.3 Guided Instruction: Multiplying Binomials

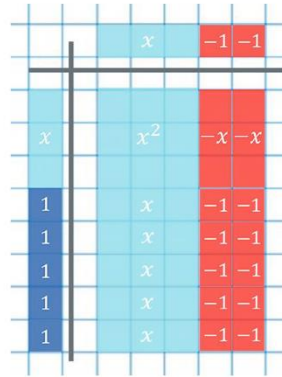
**Component Overview:** Students will be introduced to multiplying binomials using algebra tiles. For this activity, students will need modeling from the teacher during the instruction.



### 2.6.3 Guided Instruction: Multiplying Binomials

Similar to the area model, algebra tiles can be used to model the product of two binomials.

1. Ty wants to find the product of  $(x + 5)$  and  $(x - 2)$ . He uses algebra tiles to model multiplying  $(x + 5)(x - 2)$ .



Similar to the area model, the factors are placed along the width and the height of a rectangular area.

- a. Discuss with your partner how each factor is represented using algebra tiles. Write a summary of your discussion.

*Answers will vary. Each term of the factor is represented by corresponding Algebra tiles. For example,  $(x - 2)$  is represented by an 'x' tile and two  $-1$  tiles.*

- b. Complete Ty's model to find the product of  $(x + 5)$  and  $(x - 2)$  by drawing and labeling the missing tiles.

*Shown on model above.*

- c. Determine how many of each tile are represented in the product and write the numbers in the table.

| $x^2$ tiles | $+x$ tiles | $-x$ tiles | $+1$ tiles | $-1$ tiles |
|-------------|------------|------------|------------|------------|
| 1           | 5          | 2          | 0          | 10         |



For question 1, model how to use the algebra tiles to multiply the binomials using the language reviewed in the previous component and making connections to the area model.



Consider asking the following:

- Do you think that you'd still get the same product if the binomials were switched? Why or why not?



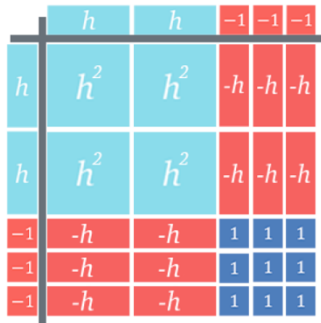
Walk students through question 1b-d. Ensure that students are positioned to hear your thinking step by step. Ask students to clarify what they think you'd do next and why.

## 2.6.3 Guided Instruction: Multiplying Binomials (continued)

- d. Complete the expression that represents the product of  $(x + 5)$  and  $(x - 2)$  using the algebra tiles.

$$(x + 5)(x - 2) = 1x^2 + 3x - 10$$

2. Monica is using algebra tiles to find the product of  $(2h - 3)^2$ .



- a. How do the algebra tiles model represent  $(2h - 3)^2$ ?

*Each side of the rectangle model represents  $2h - 3$  which is the product of  $(2h - 3)(2h - 3)$ .*

- b. Complete Monica's model.

*Answers should include: 4  $h^2$  tiles, 12  $-h$  tiles, and 9 1 tiles.*

- c. What shape did the product of  $(2h - 3)^2$  form using the algebra tiles?

*A square.*

Remember when a value is "squared," the value is being multiplied by itself.

- d. Complete the table by writing how many of each tile are in the model of the product of  $(2h - 3)^2$ .

| $h^2$ tiles | $+h$ tiles | $-h$ tiles | $+1$ tiles | $-1$ tiles |
|-------------|------------|------------|------------|------------|
| 4           | 0          | 12         | 9          | 0          |



*Some students may benefit from having physical algebra tiles available to manipulate for this activity.*



*Consider modeling question 2 as well as this can be a bit more challenging for students. The students may write this as  $(2h)^2 + (-3)^2$  instead of  $(2h - 3)(2h - 3)$ . Highlight this misconception up front before moving into the algebra tiles piece. Release students to try questions 2a-e with their partner and then come back together as a class to review.*



*After question 2c, consider asking the following.*

- Do you think the shape would remain a square if it were  $(2h - 3)(2h + 3)$ ? Why or why not?*



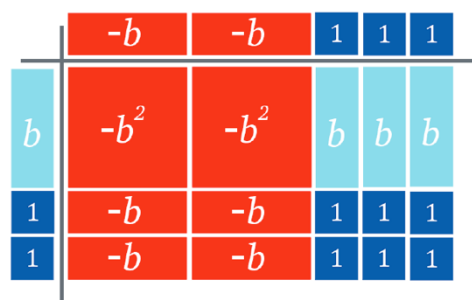
## 2.6.3 Guided Instruction: Multiplying Binomials (continued)

- e. Write the algebraic expression that represents the product of  $(2h - 3)^2$ .

$$4h^2 - 12h + 9$$

Work with your partner to complete the following.

3. Annalee used algebra tiles to find the product of  $(b + 2)(3 - 2b)$ . Her model is shown.



- a. Complete the statements.

The factor  $(b + 2)$  is represented along the

- ☐ width
- ☒ height

of Annalee's model, while the factor

$(3 - 2b)$  is represented along the

- ☒ width.
- ☐ height.

- b. Annalee wrote the product is  $2b^2 - 7b + 6$ . Describe the error(s) Annalee made.

*Annalee added the two  $-b^2$  tiles and got a positive answer instead of a negative. Annalee added the four  $-b$  tiles and the three  $b$  tiles incorrectly and should have gotten  $-4b + 3b = -b$ .*

- c. Complete the statement.

The product of  $(b + 2)(3 - 2b)$  is  $-2b^2 - b + 6$ .



*Students often miss this when multiplying using algebra tiles or area models. Ensure they are writing the product as an algebraic expression. As instruction moves from concrete to abstract in future lessons, there will be important connections that need to be made in the transition.*



*Allow informal observation throughout the first two questions to dictate how "guided" this question should be for you and your classroom. This could mean small group instruction for some while others attempt independently, or it could mean teacher modeling while students actively work as a whole group.*



*To probe deeper into the student thinking, consider asking the following after question 3b.*

- *How would you show Annalee that she made an error so that she won't make the same mistake in the future?*



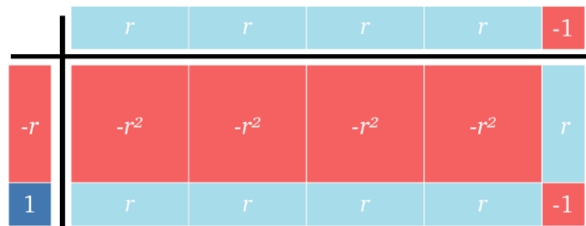
## 2.6.4 Guided Practice: Multiplying Binomials

**Component Overview:** Students will complete two practice questions on finding the product of binomials.



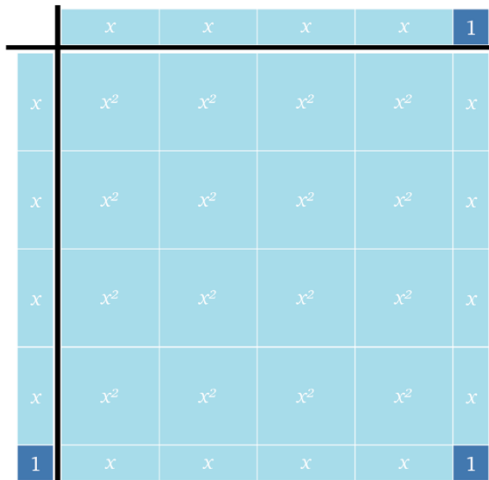
### 2.6.4 Guided Practice: Multiplying Binomials

1. Use algebra tiles to find the product of  $(1 - r)(4r - 1)$ .



$$(1 - r)(4r - 1) = \underline{-4r^2 + 5r - 1}$$

2. Find the product of  $(4x + 1)^2$  using algebra tiles.



$$(4x + 1)^2 = \underline{16x^2 + 8x + 1}$$



Use observational data from the 2.6.2 and 2.6.3 to determine how much guidance is needed in this Guided Practice component. If necessary, consider pulling a small group as they work through the questions to address common misconceptions and solidify understanding.



For students who finish early, consider adding a challenge of having them create a problem for a partner and having the partner represent it with algebra tiles.

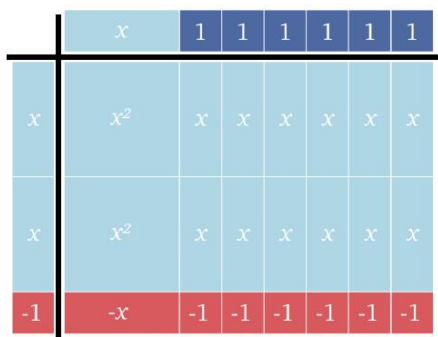
## 2.6.5 Your Turn!

**Component Overview:** Students will complete two practice questions multiplying binomials.



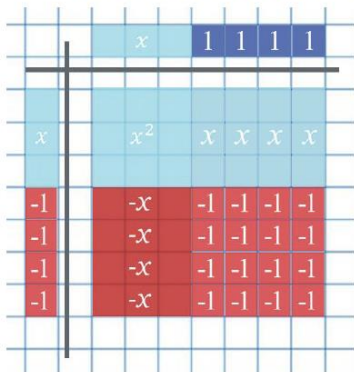
### 2.6.5 Your Turn!

1. Use algebra tiles to find the product of  $(2x - 1)(x + 6)$ .



$$(2x - 1)(x + 6) = 2x^2 + 11x - 6$$

2. Use algebra tiles to find the product of  $(x - 4)(x + 4)$ .



$$(x - 4)(x + 4) = x^2 - 16$$



For 2.6.5 Your Turn!, allow students to try these questions completely on their own without prompting and support. Once they complete them, have them check their work with a partner to determine if their answers match.

If time is an issue in this lesson, consider assigning this component for homework.

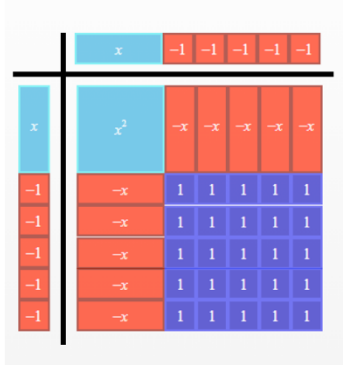
## 2.6.6 Wrap-Up: Cool Down

**Component Overview:** Students will complete a question that assesses their understanding of multiplying binomials using algebra tiles. This activity should be done independently to gather accurate student-level data.



### 2.6.6 Wrap-Up: Cool Down

- Use algebra tiles to find the product of  $(x - 5)^2$ .



$$(x - 5)^2 = x^2 - 10x + 25$$



Students may get the algebra tiles part wrong but get the product correct or vice-versa. The part(s), if any, from this question that students get incorrect can provide information on where they may need additional support.